

Geometric Numerical Integrators

The Störmer-Verlet Method and Its Properties

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Table of Contents

- 1 Motivation
- 2 Hamiltonian Systems
- 3 Störmer-Verlet Method
- 4 Symplecticity
- 5 Area Preservation
- 6 Time Reversibility
- 7 Applications
- 8 Summary

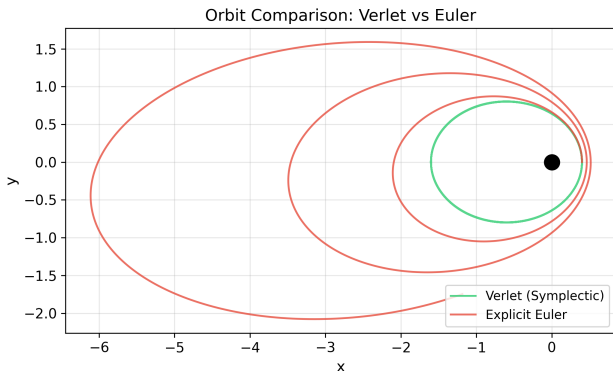
Why Do Geometric Integrators Matter?

Problems with classical methods:

- Work well in the short term
- Long term: **energy drift**
- Trajectory distortion
- Physically incorrect results

Application areas:

- Celestial mechanics
- Molecular dynamics (10^6 – 10^9 steps!)
- Vibrational mechanical models



Hamiltonian Systems

Definition

Hamiltonian systems are described by the differential equations:

$$\dot{p} = -\nabla_q H(p, q), \quad \dot{q} = \nabla_p H(p, q)$$

where $H(p, q)$ is the *Hamiltonian function* (energy).

Separable Hamiltonian

Common special case in physical systems:

$$H(p, q) = \underbrace{\frac{1}{2} p^T M^{-1} p}_{\text{kinetic energy}} + \underbrace{U(q)}_{\text{potential energy}}$$

Second-order form: $\ddot{q} = f(q)$, where $f(q) = -M^{-1} \nabla U(q)$.

Geometric Properties of Hamiltonian Systems

- ① **Symplecticity:** The flow φ_t preserves the symplectic structure:

$$\varphi'_t(p, q)^T J \varphi'_t(p, q) = J$$

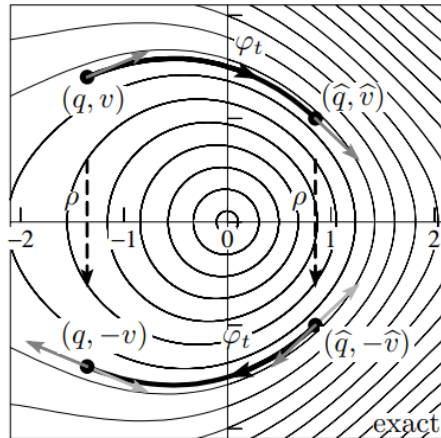
where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$

$\Rightarrow$ Area preservation (Liouville's theorem)

- ② **Energy conservation:** $H(p(t), q(t)) = \text{constant}$
 $\Rightarrow$ The Hamiltonian is a *first integral*.

- ③ **Reversibility:** Reversing time ($t \rightarrow -t$, $p \rightarrow -p$) retraces the original trajectory

Goal: Numerical methods that preserve these properties!



Derivation of the Störmer-Verlet Method

Starting point: $q'' = f(q)$ second-order differential equation

Discretization: $\frac{q_{n+1} - 2q_n + q_{n-1}}{h^2} = f(q_n)$

Introducing auxiliary variables ($v = q'$):

$$v_{n+\frac{1}{2}} = \frac{q_{n+1} - q_n}{h}, \quad v_n = \frac{q_{n+1} - q_{n-1}}{2h}, \quad q_{n+\frac{1}{2}} = \frac{q_n + q_{n-1}}{2}$$

Störmer-Verlet (A)

$$(q_n, v_n) \rightarrow (q_{n+1}, v_{n+1})$$

$$v_{n+\frac{1}{2}} = v_n + \frac{h}{2} f(q_n)$$

$$q_{n+1} = q_n + h v_{n+\frac{1}{2}}$$

$$v_{n+1} = v_{n+\frac{1}{2}} + \frac{h}{2} f(q_{n+1})$$

Störmer-Verlet (B)

$$(q_{n-\frac{1}{2}}, v_{n-\frac{1}{2}}) \rightarrow (q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}})$$

$$q_n = q_{n-\frac{1}{2}} + \frac{h}{2} v_{n-\frac{1}{2}}$$

$$v_{n+\frac{1}{2}} = v_{n-\frac{1}{2}} + h f(q_n)$$

$$q_{n+\frac{1}{2}} = q_n + \frac{h}{2} v_{n+\frac{1}{2}}$$

Properties: Second-order, explicit for $f(q)$ -type equations

Sub-operators: SE1 and SE2

Split the methods into smaller sub-operators!

SE1 method:

$$(q_n, v_n) \rightarrow (q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}})$$

$$v_{n+\frac{1}{2}} = v_n + \frac{h}{2} f(q_n)$$

$$q_{n+\frac{1}{2}} = q_n + \frac{h}{2} v_{n+\frac{1}{2}}$$

SE2 method:

$$(q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}}) \rightarrow (q_{n+1}, v_{n+1})$$

$$q_{n+1} = q_{n+\frac{1}{2}} + \frac{h}{2} v_{n+\frac{1}{2}}$$

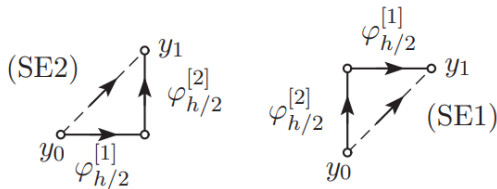
$$v_{n+1} = v_{n+\frac{1}{2}} + \frac{h}{2} f(q_{n+1})$$

Composition

The Störmer-Verlet methods:

$$(A) = (\text{SE2}) \circ (\text{SE1})$$

$$(B) = (\text{SE1}) \circ (\text{SE2})$$



Geometric Interpretation of the Method

Splitting method:

Split the vector field:

$$(v, f(q)) = (v, 0) + (0, f(q))$$

Two sub-operators:

$$\varphi_t^{[1]} : \begin{cases} q_1 = q_0 + tv_0 \\ v_1 = v_0 \end{cases} \quad (\text{drift})$$

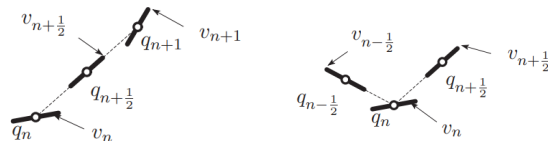
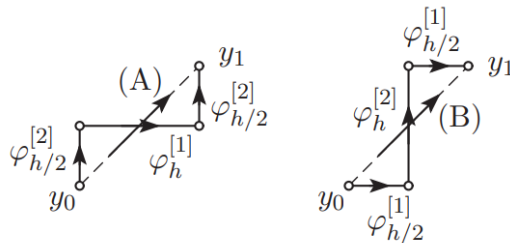
$$\varphi_t^{[2]} : \begin{cases} q_1 = q_0 \\ v_1 = v_0 + tf(q_0) \end{cases} \quad (\text{kick})$$

Störmer-Verlet (A):

$$\Phi_h^{(A)} = \varphi_{h/2}^{[2]} \circ \varphi_h^{[1]} \circ \varphi_{h/2}^{[2]}$$

Störmer-Verlet (B):

$$\Phi_h^{(B)} = \varphi_{h/2}^{[1]} \circ \varphi_h^{[2]} \circ \varphi_{h/2}^{[1]}$$



Generalization to Arbitrary Systems

Question: Can we use Störmer-Verlet for general first-order systems?

Original: $q'' = f(q)$

$$\begin{cases} q' = v \\ v' = f(q) \end{cases}$$

General: arbitrary system

$$\begin{cases} q' = g(q, v) \\ v' = f(q, v) \end{cases}$$

Methods are explicit:

- Direct computation
- Low computational cost
- Ideal for this form!

SE1 and SE2 become implicit:

$$v_{n+\frac{1}{2}} = v_n + \frac{h}{2} f(q_n, v_{n+\frac{1}{2}})$$

$$q_{n+1} = q_{n+\frac{1}{2}} + \frac{h}{2} g(q_{n+1}, v_{n+\frac{1}{2}})$$

Conclusion

Störmer-Verlet is best suited for $q'' = f(q)$ type equations!

For general systems, the implicit solves become computationally expensive.

Implicit Nature of SVA and SVB for General Systems

Both SVA and SVB have TWO implicit steps for general systems!

Störmer-Verlet (A)

For general system $q' = g(q, v)$, $v' = f(q, v)$:

$$v_{n+\frac{1}{2}} = v_n + \frac{h}{2} f(q_n, v_{n+\frac{1}{2}})$$

$$q_{n+1} = q_n + \frac{h}{2} \left(g(q_n, v_{n+\frac{1}{2}}) + g(q_{n+1}, v_{n+\frac{1}{2}}) \right)$$

$$v_{n+1} = v_{n+\frac{1}{2}} + \frac{h}{2} f(q_{n+1}, v_{n+\frac{1}{2}})$$

Two implicit solves required!

Störmer-Verlet (B)

For general system $q' = g(q, v)$, $v' = f(q, v)$:

$$q_n = q_{n-\frac{1}{2}} + \frac{h}{2} g(q_n, v_{n-\frac{1}{2}})$$

$$v_{n+\frac{1}{2}} = v_{n-\frac{1}{2}} + \frac{h}{2} \left(f(q_n, v_{n-\frac{1}{2}}) + f(q_n, v_{n+\frac{1}{2}}) \right)$$

$$q_{n+\frac{1}{2}} = q_n + \frac{h}{2} g(q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}})$$

Two implicit solves required!

Key Observation

For the special case $q' = v$, $v' = f(q)$, all implicit equations become **explicit**!

This is why Störmer-Verlet excels for separable Hamiltonian systems.

What is Symplecticity?

Definition

A numerical integrator Φ_h is *symplectic* if

$$\Phi_h'(p, q)^T J \Phi_h'(p, q) = J$$

at every point (p, q) .

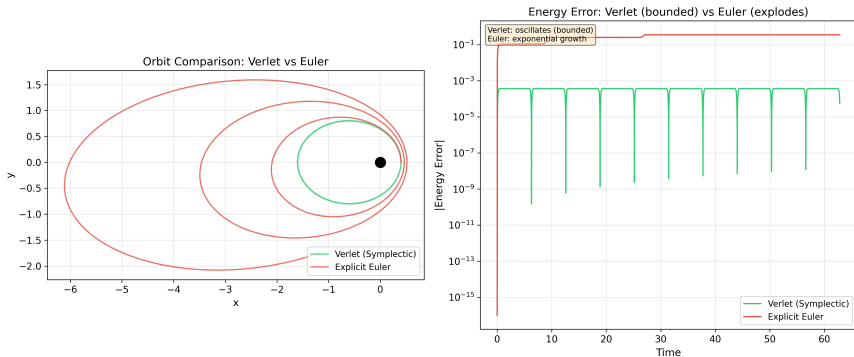
Consequences:

- Preserves phase space volume (Liouville's theorem)
- **Bounded energy error** in the long term (no drift!)
- Physically faithful simulation

Theorem: Symplecticity of Störmer-Verlet

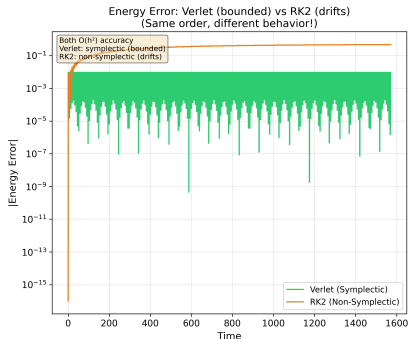
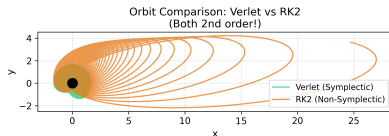
The (SE1) and (SE2) sub-operators are symplectic, therefore their composition, the Störmer-Verlet (A) and (B) methods, are also symplectic.

Störmer-Verlet vs. Explicit Euler



- **Left:** Trajectories over 10 orbits ($h = 0.01$)
- **Right:** Energy error vs. time
- **Euler:** exponential energy drift, spiral trajectory
- **Verlet:** bounded oscillating error, stable ellipse

Störmer-Verlet vs. RK2: The Key Comparison

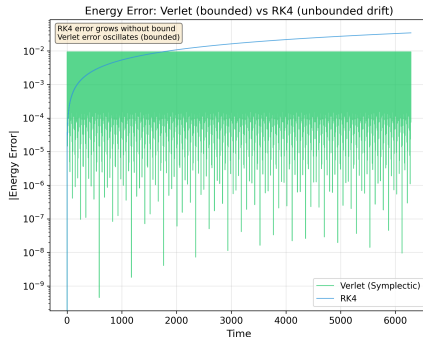
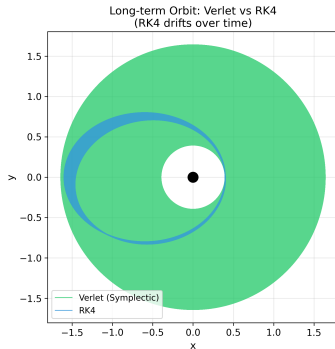


Both methods are second-order ($O(h^2)$)!

- **Verlet (symplectic):** Perfect ellipse after 250 orbits
- **RK2 (non-symplectic):** Trajectory drifts, energy grows

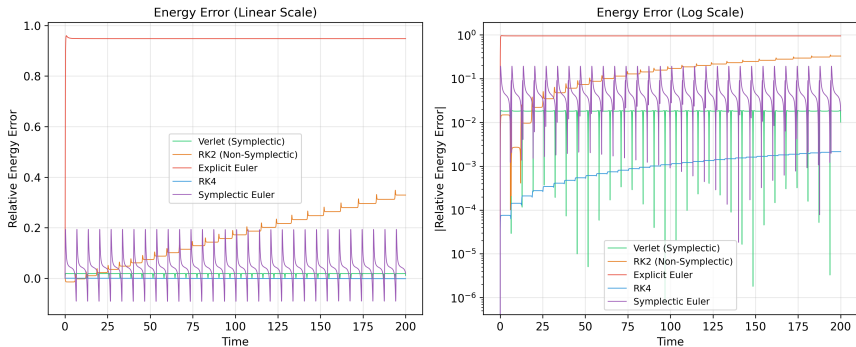
Symplecticity is more important than accuracy order!

Störmer-Verlet vs. RK4: Long-Term Behavior



- **RK4**: 4th-order accuracy, but **not symplectic**
- Short term (less than 100 orbits): RK4 is more accurate
- Long term (more than 100 orbits): **Verlet wins!**
- RK4 energy drifts, Verlet remains bounded

Energy Error Comparison: All Methods



Dichotomy:

- **Symplectic methods:** bounded oscillating error
- **Non-symplectic methods:** energy drift

Liouville's Theorem and Area Preservation

Liouville's Theorem

The flow of Hamiltonian systems preserves phase space volume:

$$\det \varphi'_t(p, q) = 1 \quad \forall t$$

For numerical methods: A method preserves area if

$$\det \frac{\partial}{\partial(p_0, q_0)} \Phi_h = 1$$

Example - Pendulum motion: $p' = q$, $q' = -\sin p$

Explicit Euler:

$$\det \Phi_h^{EE} = 1 + h^2 \cos p_0 \neq 1$$

Implicit Euler:

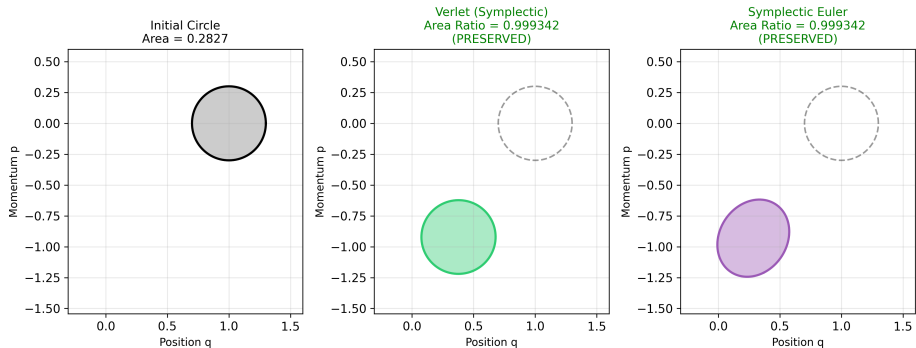
$$\det \Phi_h^{IE} = \frac{1}{1 + h^2 \cos p_1} \neq 1$$

SV (SE1, SE2):

$$\det \Phi_h^{SE1} = \det \Phi_h^{SE2} = 1 \quad \checkmark$$

Visual Demonstration: Symplectic Methods

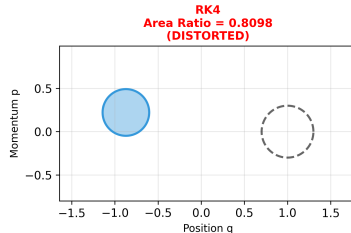
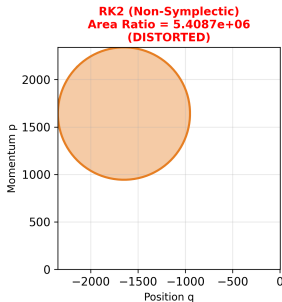
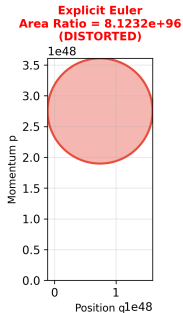
Symplectic Methods: Area is Preserved
(Circle rotates/transforms but maintains area)



- Circle of initial conditions in phase space, 100 steps ($h = 0.2$)
- **Verlet & Sympl. Euler:** Area ratio ≈ 1.000 (within 0.1%)
- The circle rotates and deforms, but **area is preserved**

Visual Demonstration: Non-Symplectic Methods

Non-Symplectic Methods: Area is NOT Preserved
(Circle expands/contracts, violating Liouville theorem)



- Same test, 1000 steps, $h = 0.5$

- **Euler:** Drastic expansion

- **RK2:** Significant distortion

- **RK4:** Measurable distortion

Symmetry and Reversibility

Time Reversibility

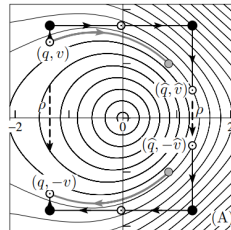
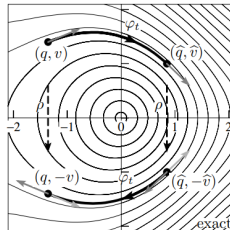
Hamiltonian systems are reversible: if we reverse time and momenta ($t \rightarrow -t$, $p \rightarrow -p$), we retrace the original trajectory.

Störmer-Verlet is symmetric:

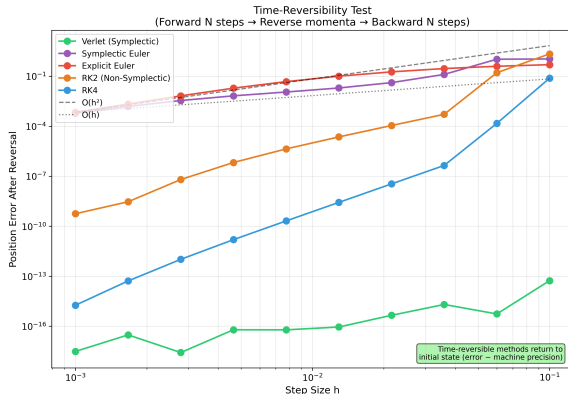
$$\begin{aligned}\text{Forward: } v_{n+\frac{1}{2}} &= v_n + \frac{h}{2}f(q_n) \\ q_{n+1} &= q_n + hv_{n+\frac{1}{2}} \\ v_{n+1} &= v_{n+\frac{1}{2}} + \frac{h}{2}f(q_{n+1})\end{aligned}$$

If we reverse: $h \rightarrow -h$, $v \rightarrow -v$, we **exactly recover** (q_n, v_n) !

Kick-Drift-Kick $\Rightarrow$ symmetry



Reversibility Test



Test: N steps forward $\rightarrow$ momentum reversal $\rightarrow N$ steps backward

- **Verlet:** Error $\approx 10^{-14}$ (machine precision!)
- Euler, RK2, RK4: Errors scale with $O(h^p)$
- Only symmetric methods are perfectly reversible

Two-Particle Lennard-Jones Simulation

Lennard-Jones potential:

Interaction of noble gases, simple molecules

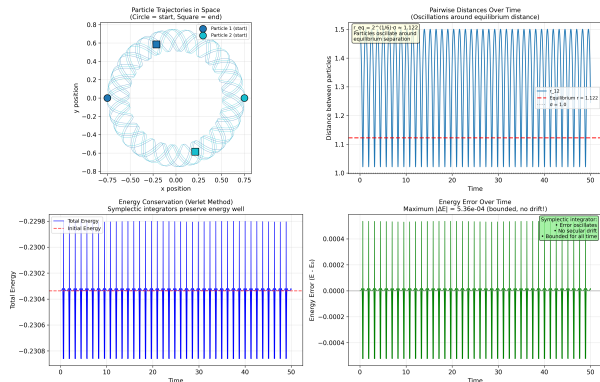
$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

- r^{-12} : Pauli repulsion
- r^{-6} : van der Waals attraction
- Equilibrium distance:
 $r_{eq} = 2^{1/6}\sigma \approx 1.122\sigma$

Simulation results:

- **Upper left:** Particle trajectories
- **Upper right:** Distance oscillates around r_{eq}
- **Lower left:** Energy converts
- **Lower right:** Relative energy error stable!

Lennard-Jones Molecular Dynamics (2 particles)



Three-Particle Lennard-Jones Simulation

Many-body system:

- 3 pairwise interactions
- Complex dynamics
- **Energy is conserved!**

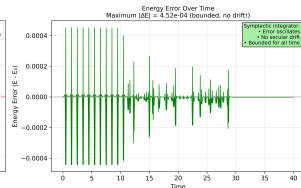
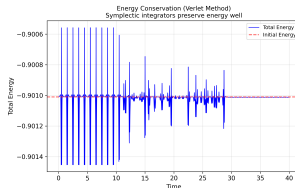
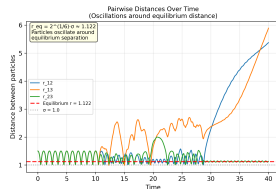
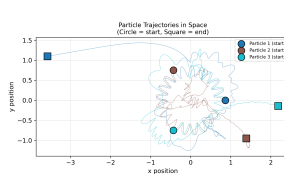
Scalability:

- Larger systems: 10^4 – 10^9 atoms
- Real MD simulations: 10^6 – 10^9 time steps
- **Verlet is the only option** for long-term stability

Industrial applications:

- GROMACS, LAMMPS
- AMBER, NAMD
- Protein dynamics
- Materials design

Lennard-Jones Molecular Dynamics (3 particles)



Comparison of Implemented Methods

Method	Type	Order	Symplectic	Reversible
Störmer-Verlet	Splitting	2	✓	✓
Sympl. Euler	Splitting	1	✓	—
Explicit Euler	Standard	1	—	—
RK2 (Midpoint)	Runge-Kutta	2	—	—
RK4	Runge-Kutta	4	—	—

Key Question

Is it worth using higher-order methods?

The answer: NOT always!

Key Message

For Hamiltonian systems:

Geometric structure $>$ Algebraic accuracy order

Advantages of Störmer-Verlet:

- **Symplectic:** Preserves phase space geometry
- **Bounded energy error:** No drift in the long term
- **Area preservation:** Liouville's theorem satisfied
- **Reversible:** Time-symmetric
- **Efficient:** Only one force evaluation per step

When to use: Conservative systems, long-term simulations

When NOT to use: Dissipative systems, very short integrations

Recommended reading:

- E. Hairer, C. Lubich, G. Wanner: *Geometric Numerical Integration* (2006)
- B. Leimkuhler, S. Reich: *Simulating Hamiltonian Dynamics* (2004)
- J.M. Sanz-Serna, M.P. Calvo: *Numerical Hamiltonian Problems* (1994)

Open-source implementation:

- GitHub repository: Python code + Jupyter notebook
- 14 visualizations + 4 animations
- Full reproducibility
- Detailed documentation (PLOTS_DOCUMENTATION.md, README.md)

Thank you for your attention!

Questions?